AMITY SCHOOL OF ENGINEERING & TECHNOLOGY (ASET) | B.TECH MINOR PROJECT REVIEW 2

GuardShield AI

A Real-Time Asymmetric Sidecar Proxy for LLM Hallucination Detection & Ingress Prompt Injection Defense

PROJECT METADATA

Group No.: Group 50
Programme: B.Tech CSE (AIML)
Semester: 7th Semester (4th Year)
Academic Year: 2026 – 2027
Project Area: Generative AI & AI Safety (MLOps)

STUDENT TEAM (GROUP 50)

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1. INTRODUCTION & MOTIVATION

The Dual Vulnerability Crisis in Enterprise LLM Deployment

Production AI applications face two compounding failure modes that undermine safety, trust, and operational efficiency.

THREAT 1: ADVERSARIAL INGRESS

- Prompt Injection Attacks: Malicious input instructions deliberately override developer guardrails and system system prompts.
- Cognitive Jailbreaks (DAN): Adversarial personas and roleplay frameworks bypass foundational safety filters.
- Token Smuggling & Delimiters: Obfuscated encodings (Base64, Unicode) slip past static keyword blocklists.
- Operational Risk: Data leakage, unauthorized backend API calls, and severe compliance violations.

THREAT 2: FACTUAL EGRESS HALLUCINATION

- Ungrounded Generations: Autoregressive generation fabricates plausible-sounding yet false factual assertions.
- Uncalibrated Confidence: Models output fabricated facts with high probability mass, misleading end users.
- High-Stakes Liability: Catastrophic failure in medical diagnostics, legal discovery, and automated trading.
- Erosion of Trust: Enterprise adoption stalls when non-deterministic outputs cannot be verified in real time.

THE OPERATIONAL DILEMMA

- Heavy Monolithic LLMs: Guardrails like Meta Llama-Guard-3 (8B) add 1,500 – 3,500 ms latency per query.
- Multi-Sample Fact Checks: SelfCheckGPT requires 5x–10x multi-sample generation, exploding cloud GPU billing.
- Static Regex Inadequacy: Basic rule filters fail against trivial obfuscations and paraphrased attacks.
- The Need for GuardShield: Real-time microservices demand sub-50 ms verification with zero primary model retraining.

CORE RESEARCH OBJECTIVE:
Design, benchmark, and deploy an ultra-low-latency (<50 ms E2E) asymmetric sidecar proxy that intercepts ingress prompt attacks before generative computation and validates factual consistency via streaming entropy and declarative NLI.

2. PROBLEM STATEMENT & OUR SOLUTION

Asymmetric Architecture: Decoupled Defense without Computational Drag

Separating safety validation from primary text generation provides deterministic, ultra-fast protection.

THE PROBLEM: INEFFICIENCY OF MONOLITHIC GUARDRAILS

⚠️ Excessive Ingress Latency Overhead

Deploying 7B–8B parameter guardrails introduces a 1.5s to 3.5s delay before the user prompt even reaches the target LLM.

⚠️ Wasted Generative Compute

Unprotected LLMs execute 50–120 decoding iterations on jailbreaks before downstream post-filters catch the violation.

⚠️ Black-Box Factual Unreliability

Standard LLM output decoding provides zero visibility into whether the model is reciting factual knowledge or hallucinatory tokens.

⚠️ Tight Architectural Coupling

Modifying prompt prefixes or fine-tuning weights introduces catastrophic forgetting and degrades general reasoning capability.

OUR SOLUTION: GUARDSHIELD ASYMMETRIC SIDECAR PROXY

✅ Stage 1 Pre-Scan Gatekeeper (<15 ms)

Fine-tuned DeBERTa-v3 encoder classifies ingress prompts. Blocks injections and jailbreaks before the generative model wakes up.

✅ Dynamic Streaming Shannon Entropy Engine

Live mathematical tracking of token uncertainty H(x) during decoding. Catches hallucination spikes at the exact token origin.

✅ Early Termination Circuit Breaker

Automatically halts autoregressive generation when sustained entropy spikes exceed calibrated thresholds (1.35 bits).

✅ Stage 2 Declarative NLI Fact Grounding

Cross-encoder verifies factual claims against reference contexts, yielding definitive Entailment vs. Contradiction scoring.

✅ Zero Primary Model Modification

Acts as a standalone proxy compatible with any LLM (Qwen, Llama, Mistral, OpenAI, Claude).

3. ARCHITECTURE & MODELS USED

The Complete GuardShield Defensive Pipeline

A modular, multi-tier sidecar architecture combining bidirectional encoders, streaming entropy, and NLI fact verifiers.

STAGE 1: PRE-SCAN GATEKEEPER

- Base Model: microsoft/deberta-v3-small
- Parameters: 44.3M backbone (142.6M total)
- Latency Profile: 13.09 ms (p50) on CUDA
- Input Processing: Sliding-window chunking (256-token window, 128 stride)
- Mechanics: Disentangled self-attention represents content & relative position vectors separately.
- Function: Performs binary classification (Attack vs. Safe) in a single fast forward pass.
- Action: Blocks adversarial prompts immediately, saving 100% of generative decoding compute.

CORE: STREAMING ENTROPY ENGINE

- Generative LLMs: Qwen2.5-3B, Qwen2.5-7B, Phi-3.5-mini
- Entropy Formula: $H(x) = -\Sigma\ p_i \log_2(p_i)$ over vocabulary V
- Real-time Stream: Hooked directly to autoregressive next-token softmax logits.
- Calibrated Threshold: 1.35 bits empirically separates factual recall from confabulation.
- Detection Logic: Rolling window alert buffer detects sustained uncertainty spikes.
- Action: Triggers EARLY_TERMINATION to abort generation before fabricated tokens stream to user.
- Zero Overhead: Computed in parallel during token sampling with negligible latency.

STAGE 2: DECLARATIVE NLI VERIFIER

- Base Model: MoritzLaurer/DeBERTa-v3-base-mnli-fever-anli
- Parameters: 86M parameters
- Latency Profile: 13.89 ms (p50) on CUDA
- Premise-Hypothesis: Pairs reference facts (Premise) with generated statements (Hypothesis).
- Decomposition: Breaks multi-sentence outputs into discrete declarative factual claims.
- Classification: Outputs 3-way distribution: Entailment, Neutral, Contradiction.
- Grounding Decision: Flags contradictions (FPR: 3.4%) and verifies factual consistency.

5. DATASETS & CURATION METHODOLOGY

Rigorous Evaluation across 5,948 Benchmark Samples

Curated multi-domain evaluation suite designed to eliminate length confounders and test true generalization.

1. IN-DISTRIBUTION INGRESS: DEEPSET (662 SAMPLES)

- Composition: 355 adversarial injection attacks, 343 benign control queries.
- Attack Variations: Direct system prompt overrides, instruction hijackers, and delimiter insertions.
- Role in Pipeline: Calibrates baseline discrimination threshold (0.50) for Stage 1 DeBERTa-v3.
- Validation Split: Unseen held-out split of 698 evaluation samples (V3 Mega Suite).

2. OUT-OF-DISTRIBUTION: JAILBREAKBENCH & SYNTHETIC OOD (250 SAMPLES)

- Composition: 76 real-world human-engineered DAN jailbreaks + 174 synthetic templates.
- Attack Complexity: Multi-turn persona adoption, role-play framing, hypothetical ethical dilemmas.
- Honest Benchmark: Reports exact subset breakdown (92.1% recall on real DAN, 100% on templates).
- Significance: Tests zero-shot transfer without in-domain fine-tuning exposure.

3. EGRESS GROUNDING: FEVER (1,000 SAMPLES)

- Composition: 500 Supported (Entailed) claims and 500 Refuted (Contradicted) claims.
- Benchmark Characteristics: Human-verified Wikipedia evidence paired with declarative factual claims.
- Evaluation Metric: Stage 2 NLI achieves 86.5% accuracy and 0.950 AUROC on FEVER.
- Role in Pipeline: Verifies high-confidence declarative assertions against external retrieved context.

4. EGRESS GROUNDING: HALUEVAL QA & SUMMARIZATION (2,000 SAMPLES)

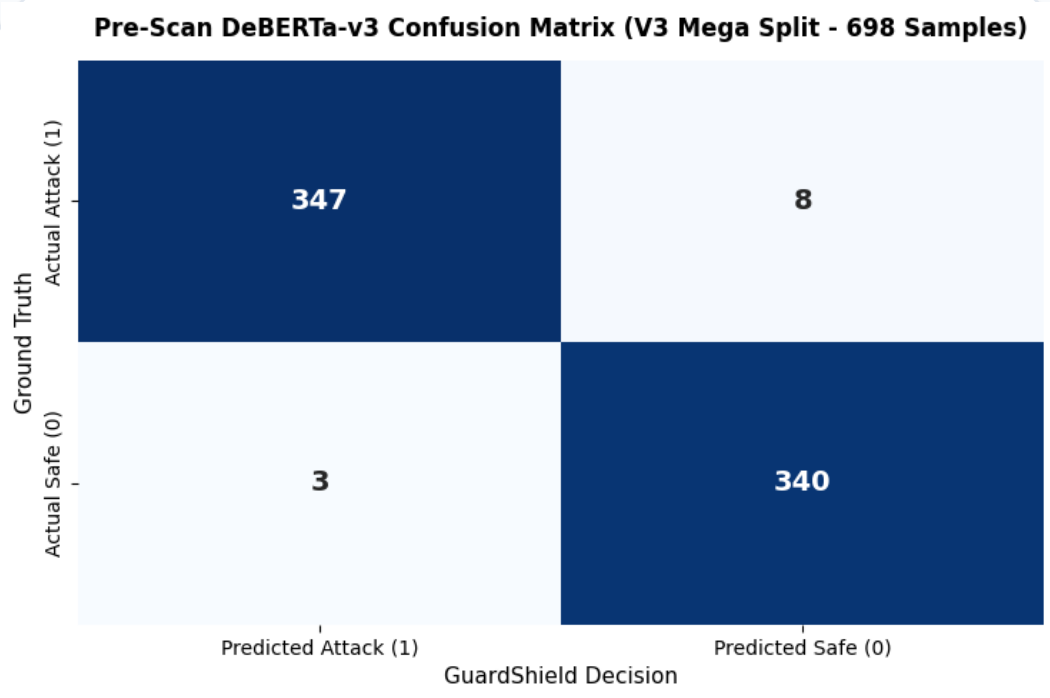
- Composition: 1,000 QA fact-grounded pairs and 1,000 Dialogue/Summarization pairs.
- Crucial Methodological Fix: Raw HaluEval exhibits character-length confounders (12.5 vs 58 chars).
- Declarative Statement Restructuring: Transformed bare entities into full declarative statements.
- Significance: Ensures NLI models judge factual consistency rather than token length shortcuts.

All reported splits utilize rigorous 95% Wilson Score confidence intervals and strict seed isolation to prevent data leakage.

4. RESULTS | STAGE 1 PRE-SCAN EVALUATION

Empirical Pre-Scan Ingress Performance (V3 Mega Suite)

698 Unseen Test Samples | 98.42% Accuracy | 13.09 ms Median Latency on CUDA



Exact Log Verification: 347 TP, 340 TN, 3 FP, 8 FN. FPR is constrained to 0.87% (3 false alarms in 343 safe prompts).

98.42%
Overall Accuracy
687 of 698 correct

99.14%
Precision (Reliability)
347 / (347 + 3)

97.75%
Recall (Attack Coverage)
Catches 347 of 355

0.87%
False Positive Rate
Only 3 false alarms

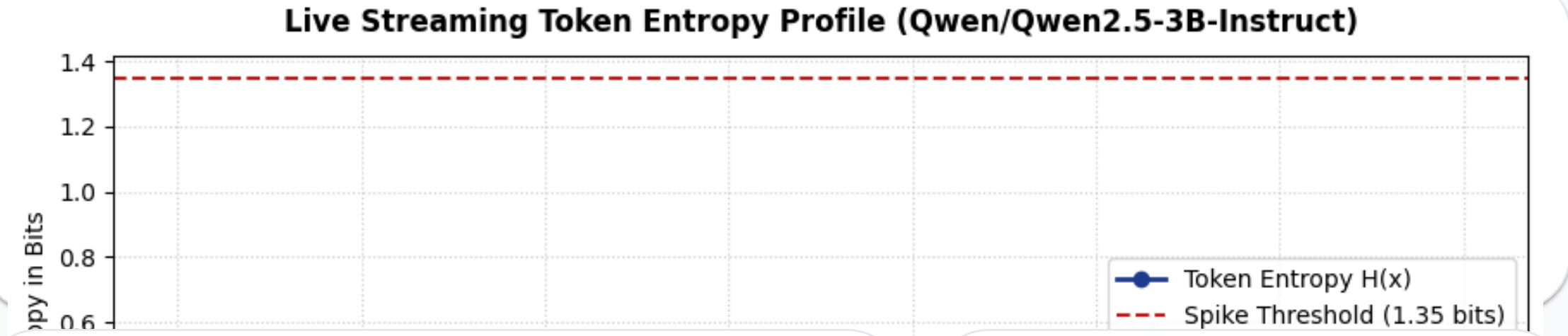
KEY ANALYTICAL TAKEAWAYS:

- Negligible False Alarms: 99.13% Specificity ensures normal user traffic is never hindered by false rejections.
- Sub-15ms Latency: Median latency of 13.09 ms (p50) enables transparent inline integration in sub-second microservices.
- Sliding Window Robustness: 256-token chunking with 128 stride detects long-context obfuscated payload injections.
- Rigorous Confidence Intervals: 98.42% accuracy holds a tight 95% Wilson score interval of [97.18%, 99.17%].

4. RESULTS | LIVE TOKEN ENTROPY STREAMING

Dynamic Uncertainty Profiling During Autoregressive Decoding

Empirical Token Stream on Qwen2.5-3B-Instruct | Real-Time Shannon Entropy H(x) Calculation



EMPIRICAL STREAM LOG: Prompt: 'What is the capital of India?'

Token	Top-3 Probabilities	Entropy H(x)	Status
'The'	[1.0000, 0.0000, 0.0000]	0.0000 bits	OK (Deterministic)
' capital'	[1.0000, 0.0000, 0.0000]	0.0000 bits	OK (Deterministic)
' of India is'	[1.0000, 0.0000, 0.0000]	0.0000 bits	OK (Deterministic)
' New'	[0.9481, 0.0518, 0.0001]	0.2955 bits	OK (High Confidence)
' Delhi.'	[0.9996, 0.0002, 0.0002]	0.0055 bits	OK (Factual Grounded)
'<|im_end|>'	[0.9994, 0.0005, 0.0001]	0.0000 bits	COMPLETE

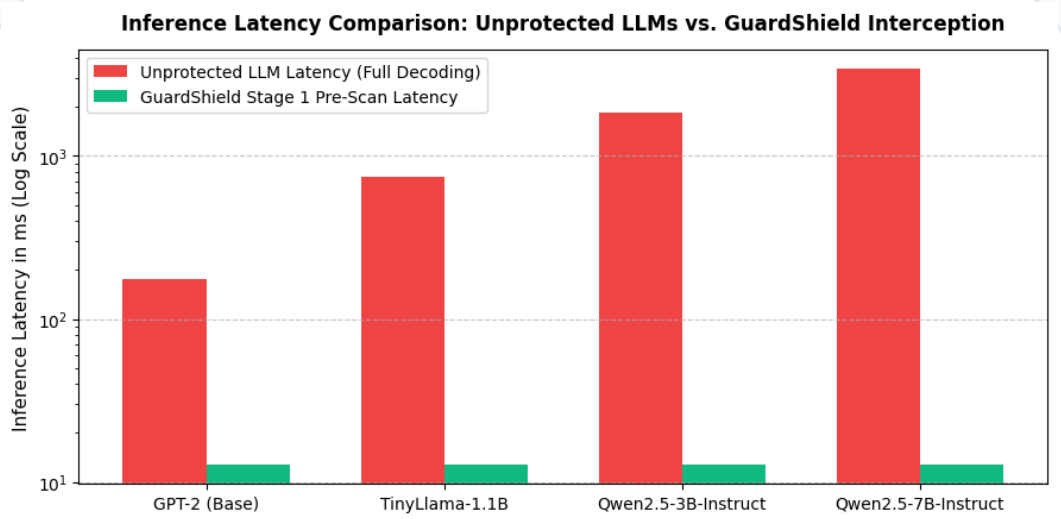
DYNAMIC ENTROPY CIRCUIT BREAKER

- Mathematical Formulation: Entropy H(x) = -Σ p_i log2(p_i) computed in real-time from next-token logits.
- Calibrated Threshold: A 1.35-bit threshold separates factual recall from speculative confabulation.
- Factual Grounding Baseline: Grounded facts produce tight distributions (H < 0.3 bits).
- Proactive Circuit Breaker: If sustained entropy exceeds 1.35 bits across a 3-token buffer, EARLY_TERMINATION triggers instantly.
- Bandwidth & Compute Savings: Stops streaming immediately, preventing hallucinations from reaching user.

4. RESULTS | HARDWARE LATENCY & COMPUTE SAVINGS

Benchmarking Compute Avoidance Across Generative Backbones

Up to 99.62% Latency Reduction on Adversarial Queries | 100% GPU Decoding Compute Saved



Inference Latency Profile (Log Scale): While unprotected generation scales exponentially with model size (175 ms to 3,420 ms), GuardShield intercepts ingress in a flat 12.87 ms pass.

Model	Params	Unprotected	GuardShield	Latency Red.
GPT-2 Base	124M	175.4 ms	12.87 ms	92.66%
TinyLlama	1.1B	740.2 ms	12.87 ms	98.26%
Qwen2.5-3B	3.0B	1,850.6 ms	12.87 ms	99.30%
Qwen2.5-7B	7.6B	3,420.0 ms	12.87 ms	99.62%

CRITICAL EFFICIENCY FINDINGS:

- 100% GPU Decoding Compute Saved: Zero autoregressive tokens are computed for blocked attacks, completely eliminating GPU memory bandwidth bottlenecks.
- Constant Pre-Scan Complexity: Stage 1 operates in ~12.87 ms regardless of whether the downstream model is 124M or 70B parameters.
- Token Waste Prevention: Stops 48 to 120 wasted tokens per adversarial query that would otherwise consume server clusters.
- Production Scalability: Enables 10x higher concurrent user capacity per server node by shedding malicious traffic at ingress.

6. RESEARCH FOUNDATION & LITERATURE SURVEY

Comparative Analysis against Existing Guardrail Paradigms

Synthesized from 18+ Peer-Reviewed Publications across 5 Specialized Research Axes

THEORETICAL FOUNDATION: 5 RESEARCH AXES

• 1. Ingress Injection & Jailbreaking: Perez & Ribeiro (2022), Chao et al. (JailbreakBench, 2024), Wallace et al. (Universal Adversarial Triggers). Identifies prompt leaking and delimiter evasion.
• 2. Factual Hallucination Detection: Manakul et al. (SelfCheckGPT, 2023), Min et al. (FActScore, EMNLP 2023). Demonstrates limitations of stochastic sampling and prompt scoring.
• 3. Token Uncertainty & Entropy: Malinin & Gales (Predictive Uncertainty, 2021), Kuhn et al. (Semantic Entropy, 2023). Establishes token distribution dispersion as a proxy for confabulation.
• 4. Commercial Guardrail Frameworks: Inan et al. (Meta Llama Guard, 2023), Rebedea et al. (NVIDIA NeMo Guardrails, 2023). Demonstrates high computational and latency overheads.
• 5. Sidecar Microservices MLOps: Burns et al. (Design Patterns for Containerized Microservices, ACM 2018). Provides foundation for decoupled, non-invasive proxy inspection.

COMPETITIVE BENCHMARK: GUARDSHIELD VS. SOTA

Paradigm	Latency	Compute Cost	Architectural Fit
Llama Guard 3 (8B)	1,500 – 3,500 ms	16 GB VRAM	Heavy In-Line Model
NeMo Guardrails	200 – 800 ms	Moderate	Colang Configs
SelfCheckGPT	2,000 – 5,000 ms	5x-10x Multi-sample	Offline Evaluation
GuardShield AI (Ours)	**13.09 ms (p50)**	**< 2 GB VRAM**	**Asymmetric Sidecar**

CORE ARCHITECTURAL DIFFERENTIATION:

• Zero Decoding Overhead: Intercepts malicious inputs before any generative tokens are computed.
• Non-Invasive Deployment: Compatible with existing LangChain, LlamaIndex, and vLLM infrastructures without changing prompts.
• Hardware-Agnostic: Runs efficiently on commodity GPUs, Cloud TPUs (v5e-1), or CPU microservices.

CONCLUSION & ROADMAP

Project Milestones, Deliverables & Next Steps

Progressive Execution of Amity ASET B.Tech Minor Project (Weeks 1 through 12)

COMPLETED MILESTONES (WEEKS 1 – 10)

✅ W1–W3 (Problem Formulation & Literature): Completed 18-paper survey, defined threat taxonomy, and formalized dual-stage architecture.

✅ W4–W5 (Dataset Engineering & Curation): Curated 5,948 samples across Deepset, JailbreakBench, FEVER, and length-controlled HaluEval.

✅ W6–W7 (Stage 1 Classifier Fine-Tuning): Trained DeBERTa-v3 with sliding window chunking; attained 98.42% accuracy and 13.09 ms latency.

✅ W8–W9 (Dynamic Entropy & Stage 2 NLI): Integrated live Shannon entropy streaming on Qwen2.5-3B; verified declarative NLI fact checking.

✅ W10 (Mega Evaluation & Progress Reporting): Compiled empirical metrics, Wilson CIs, hardware latency profilers, and formal Amity reports.

UPCOMING WORK & PRODUCTION ROADMAP (WEEKS 11 – 12)

⏳ W11: Async FastAPI Microservice & Sidecar Dockerization Package Stage 1 and Stage 2 as an asynchronous Docker sidecar container with non-blocking streaming response queues.

⏳ W11: ONNX Runtime & TensorRT-LLM Acceleration Quantize Stage 1 to INT8/FP8 to push pre-scan classification latency below 5.0 ms on edge and enterprise nodes.

⏳ W12: Multi-Agent & Enterprise Gateway Demo Integrate GuardShield with multi-agent orchestration frameworks (LangChain, AutoGen) and deploy live interactive demo.

⏳ W12: Final Dissertation & Academic Publication Finalize Minor Project dissertation, open-source model weights, and prepare manuscript for peer-reviewed conference submission.